Politicians: With Age Comes Experience

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Project Goals & Motivation

Our major goals for this project include:

- To visualize the longevity of Congress members' careers in office
- To compare the age range of members of Congress

Our motivation:

- We were curious as to how long baby boomers have maintained their positions in office, especially since we're growing up in a country where those running it have been in power for many years. This lead to a desire to graph the age range of politicians in power and how long they have been in office and show others how old our Congress really is.

Related Work

We were inspired by Hank Green's Youtube video "Why Was Every President Born in the Same Year?", which suggests a trend in politics where the age of people in power is increasing overtime. We wanted to visualize this to see how much of Congress is ruled by older citizens that have maintained power for a while.

We were also inspired by the visualizations on pudding.cool. Their pages tell a story with data and often have multiple charts that strengthen their point, and we wanted to go in a similar direction.

Questions We Want To Answer

- How many members of Congress are on the older side?
- Has the spread of ages always been the same, or is this a recent development?
- How long do Congress members remain in power?

Data

We considered projects.propublica.org at one point, but we found out that the API is no longer in use. They recommended using congress.gov instead.

Initially, we thought that the congress.gov API did not include the birth year. However, after further inspection, we were able to calculate the end year of congress by using a few endpoints (plus we got year of birth). Once that was resolved, we decided to use congress.gov as the source for all the data shown in our visualization.

To get our data, we do a query on /members, retrieving 5.5k members so that we can get the bioguideId - having a limit of 250 per query. Then we use that on /member/{bioguideId} filter to get just the general data (year of birth, terms served, etc.). This all goes into members.json. This is not served directly to the user, as it contains too much information compared to what we need. The relevant data for every year our site covers is stored via local caching so the same information isn't retrieved from the API over and over again.

Exploratory Data Analysis

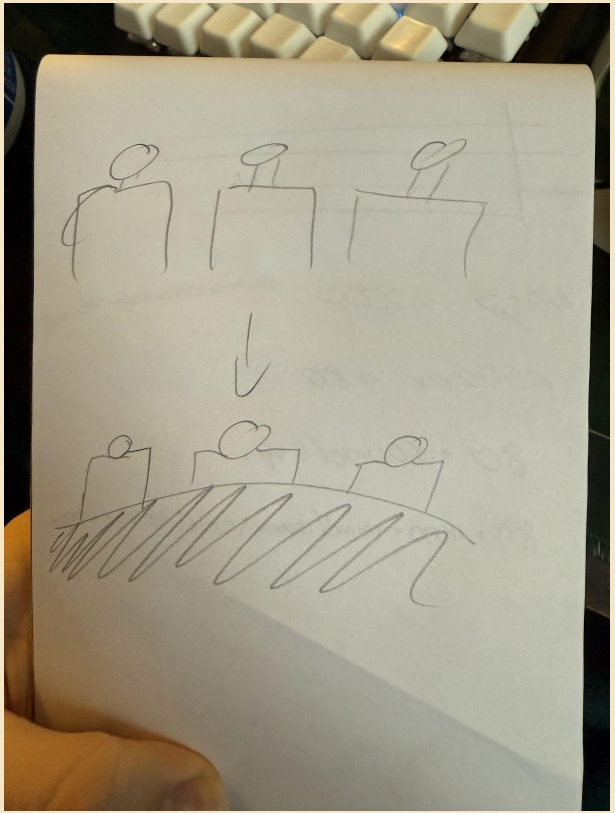
We started with a few different visualization ideas. The first is similar to an icon array, but to make it more striking, we'd have the icons for each member in curved rows similar to how an actual chamber in the Capitol building looks.

Then, to further support our point with graphs that are more intuitive to understand from a glance, we decided on a bar chart to show how many members of Congress are in different age ranges. We also chose to create a clustered bar chart using the same age brackets to show how many years members of each bracket have served in Congress on average.

Each of these visualizations would be based on the data for a year selected by the user.

Design Evolution: Congress Icon Array

Much discussion went in to how best design the figures in our icon visualization. The two data points we wanted to stress the most were the age of each member and how long they've served. From the start, we were intent on representing the age of a member based on the color of their hair. We planned that the scale would range from blonde hair representing the youngest members to white hair for the oldest members. We then came up with the idea to depict the years a member served using badges attached to each person's icon.

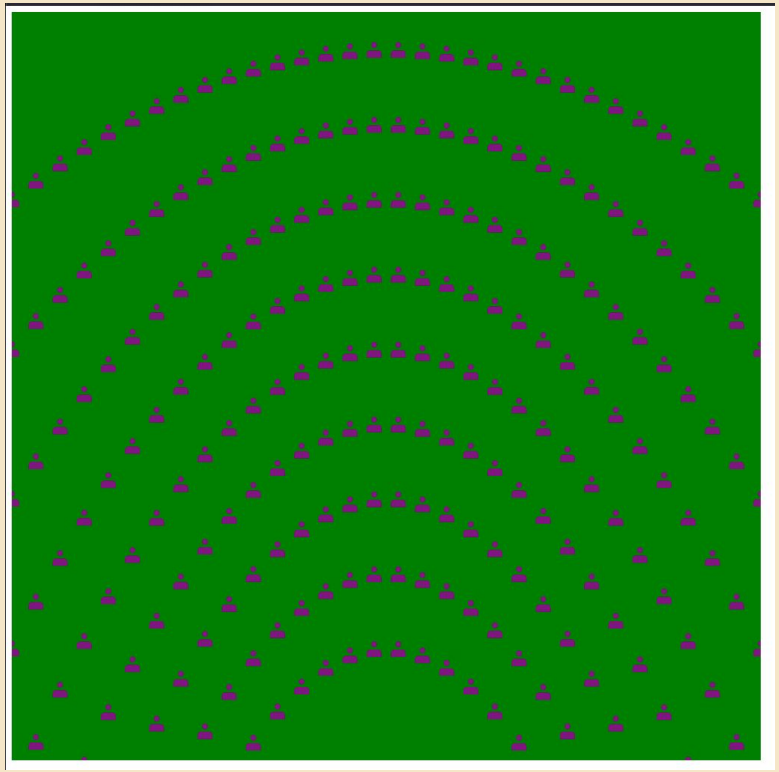


Design Evolution: Congress Icon Array

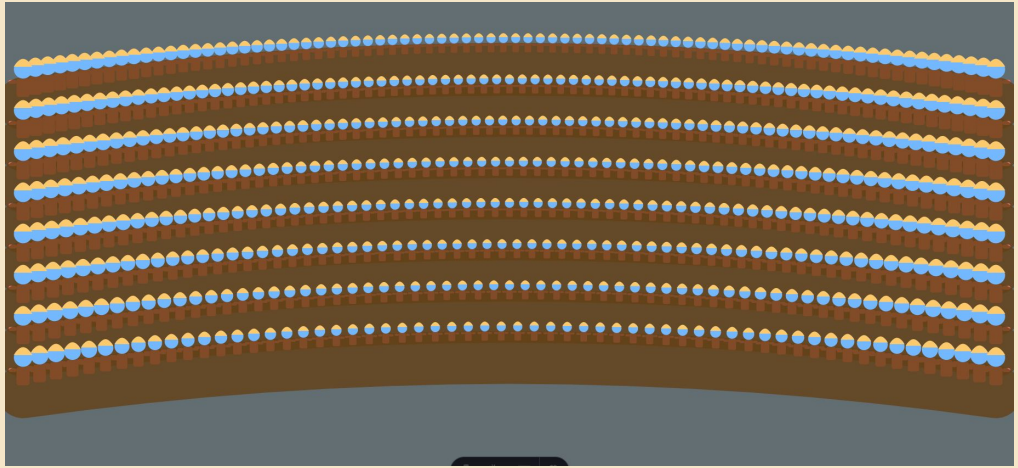
We also discussed what we didn't want to show or what we didn't think was relevant. We made a specific point not to highlight what party each member is part of, as the purpose of our visualization isn't to call attention to one party or the other, but rather highlight the aging population of Congress as a whole. We also didn’t include the gender of each member as that information was not provided by the database, and we did not find it to be vital enough to warrant finding it through other sources.

Design Evolution: Congress Icon Array

To start, we wanted a proof of concept of our icon visualization. This would confirm that we could draw enough icons to represent every member of Congress in a year while also having members in curved rows like they're sitting in a chamber. It took a bit of trial and error, with our first attempt built off of estimation quadratics in Desmos, and half of our icons in this attempt were originally being drawn outside of our SVG.



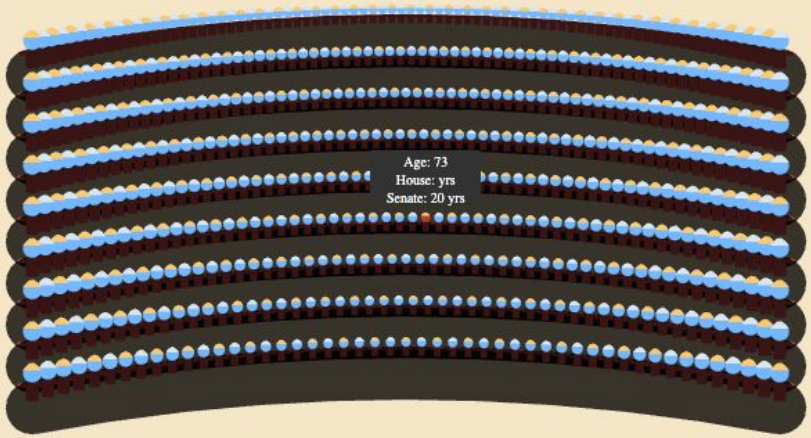
Design Evolution: Congress Icon Array



After creating a draft of a custom person icon that we'd be able to adjust depending on each member's experience, we adjusted the code to set positions of each icon. This would allow us to fit enough icons to represent each member of Congress in one visualization while still maintaining our idea of a chamber-like arrangement.

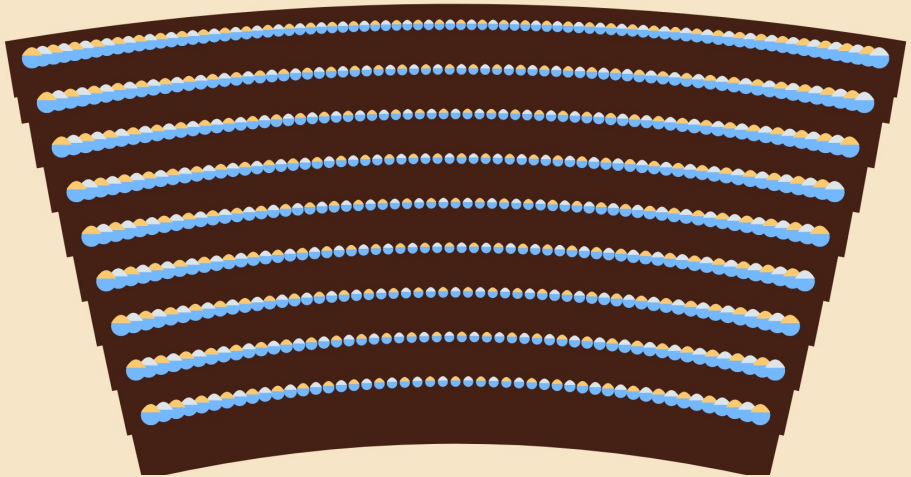
Design Evolution: Congress Icon Array

As we made more progress on our visualization, we developed the code to hover over a member to see their age and experience. At this point, the data upon hover was hard-coded and the hair color of each icon was set to alternate between each, as each icon wasn't associated with data yet but we wanted the functionality.



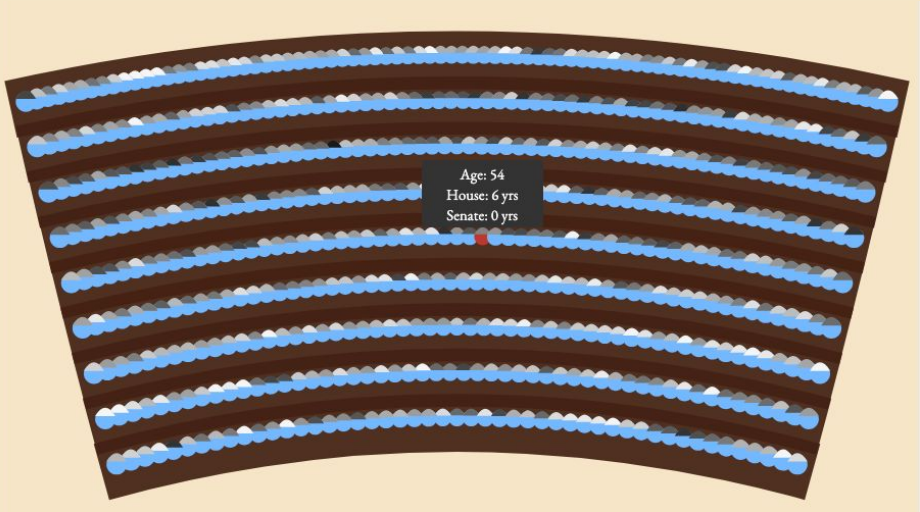
Design Evolution: Congress Icon Array

As we iterated upon the design of this visualization, we improved the shape of each row to more closely mimic a row of seats, and we decided to get rid of the bodies of our person icons entirely. We realized that with over 500 icons there was already a lot of information in the visualization at once, and adding something underneath each head to signify experience would make it much too cluttered. Instead, we visualized the experience of Congress members using a chart further down on our site.



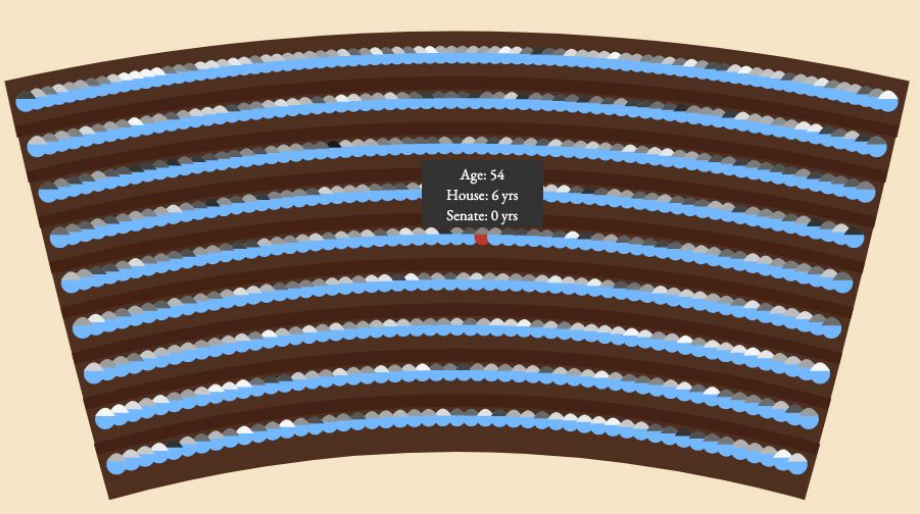
Design Evolution: Congress Icon Array

After another iteration, we made the difference in size of icons a lot more subtle, as before, it was difficult to make out the hair color of the icons in the middle. Each icon was also squished together a little more, as this allowed us to make them slightly bigger, and this choice didn't compromise the usefulness of our visualization. We also implemented the functionality of each icon getting their data from a member, and each one would have the correct info for one member upon hovering over it.



Design Evolution: Congress Icon Array

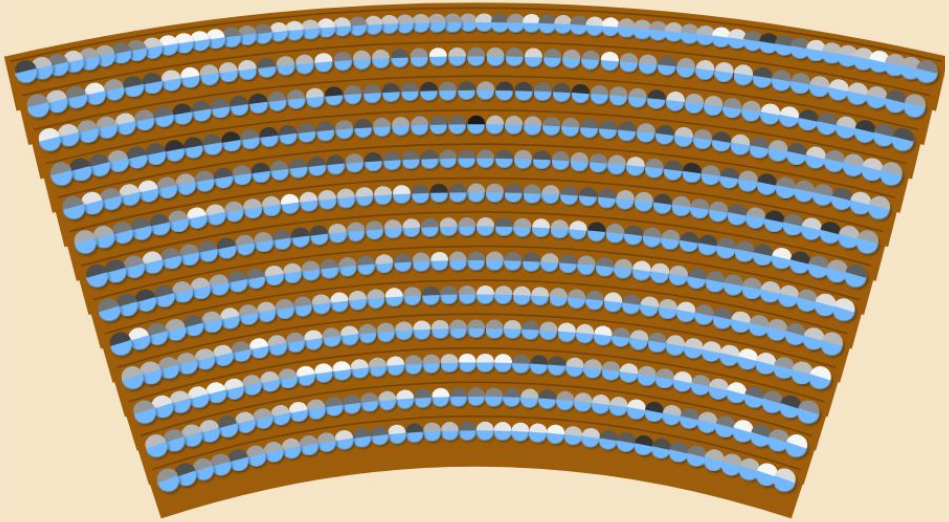
Now that we had the age of each member icon represented, we could assign the hair color of each based on a scale. After implementing this, we realized that a scale of black hair to white hair made the difference in age much more apparent. We changed up from our initial plan to make younger members blonde, and now they were black haired instead.



Design Evolution: Congress Icon Array

With the visualization receiving data properly and our hover functionality working, we made some more improvements to the graph. We made the color of the rows a light brown so that icons all across the hair-color gradient are noticeable, and we gave each icon a subtle shadow so that they pop out from the rows behind them.

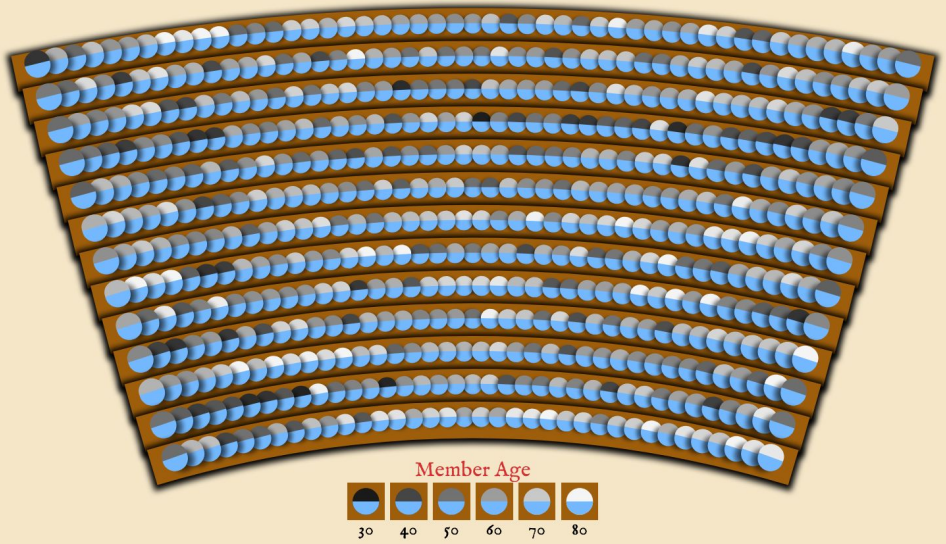
We also simplified our icons even further, making them a single circle that's half blue and half their "hair color". It's abstracted enough that our visualization can fit hundreds of them without feeling overwhelming, yet still makes it clear that it's representing people of differing ages.



Design Evolution: Congress Icon Array

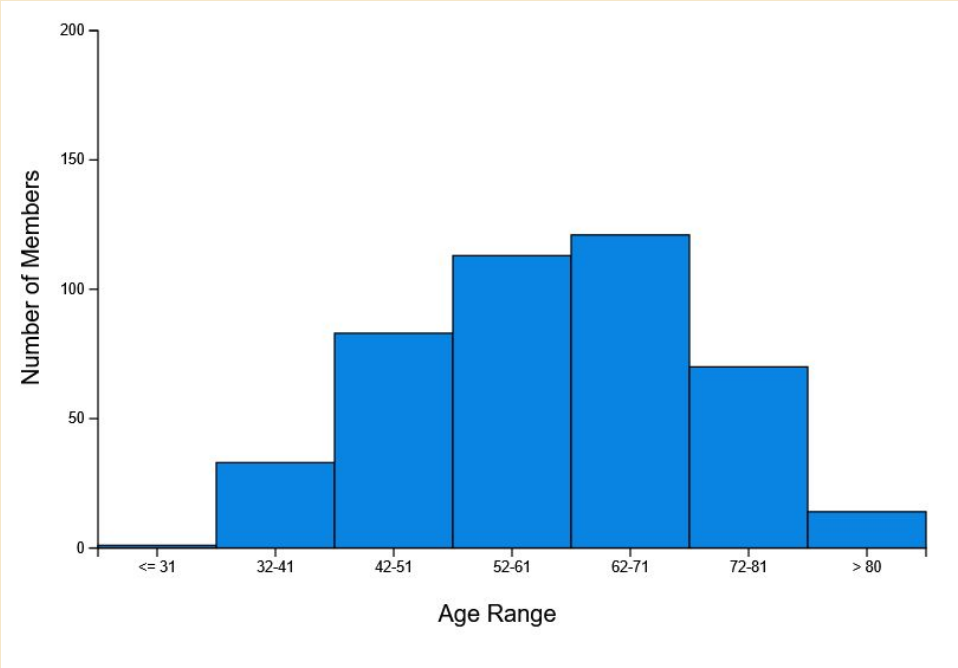
As our visualization was approaching its final design, we gave each row their own shadow and pushed the members closer together to the point of overlapping. This gives our representation of Congress more of a packed feel, emphasizing how many members there are. We also have each icon's hairline tangent to the curve they're at, working with the shadowing to build a pseudo-3D effect.

We also added a legend at the bottom so that you can see what the general age range of each member is without needing to hover over them.



Design Evolution: Age Comparison Bar Chart

Our age comparison bar chart didn't need any sketches to start out. While we planned to have two charts side by side to show membership of both the House and Senate, we started by getting one working to ensure visualization of our data worked smoothly with our code.

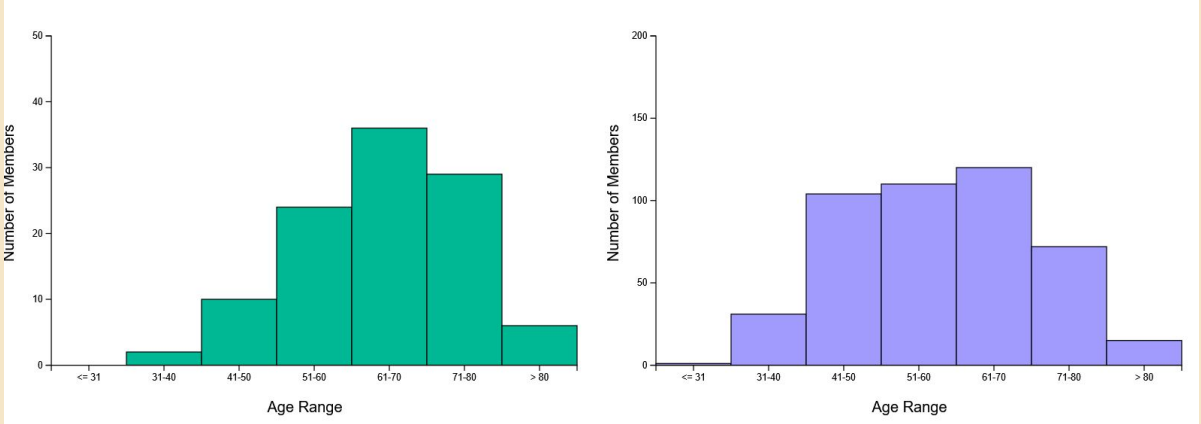


Design Evolution: Age Comparison Bar Chart

During our work on this bar chart, we noticed that the data for each Congress rarely had exactly the number of members that should be on the House and Senate, and was almost always a few members higher. Upon looking into this, we found that the data was good and the reason was because of Congress members taking over a previous member's seat while the term is ongoing. This can be due to a number of reasons, including death, resignation, promotion to another position, or other extenuating circumstances.

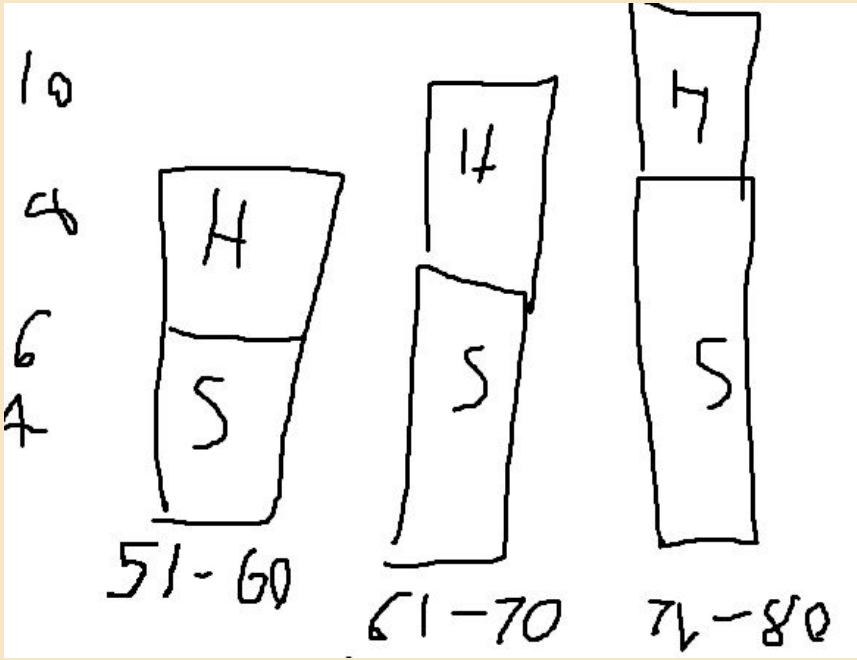
Design Evolution: Age Comparison Bar Chart

Once our bar chart code was complete and we implemented the ability for its data to change upon the user selecting another year, we put two on our page: one for the House, and one for the Senate. When we chose contrasting colors for the two chambers, we made the conscious choice not to use red or blue. Since those two colors have strong associations to parties in American politics, we avoided them so people wouldn't make any party-based assumptions about our charts.



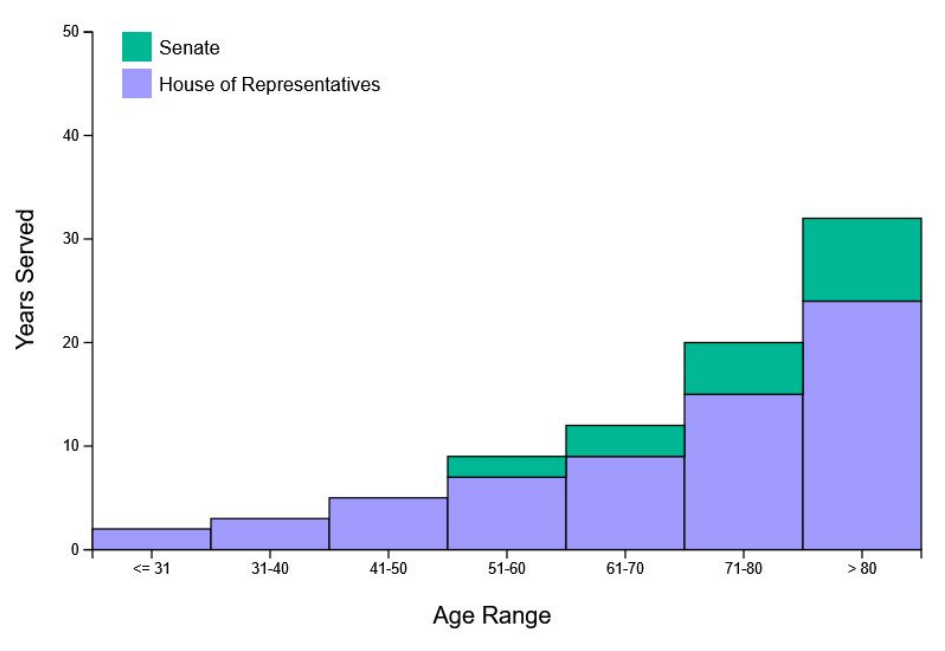
Design Evolution: Years Served Clustered Bar Chart

Besides our bar charts to show the ages of members of a current year, we also wanted to visualize not just how old members are, but how long they've been part of Congress. This would represent if the members of Congress generally remain stagnant over time or not. We started out with a simple sketch to brainstorm the idea, showing the average years served in both chambers for each age range.



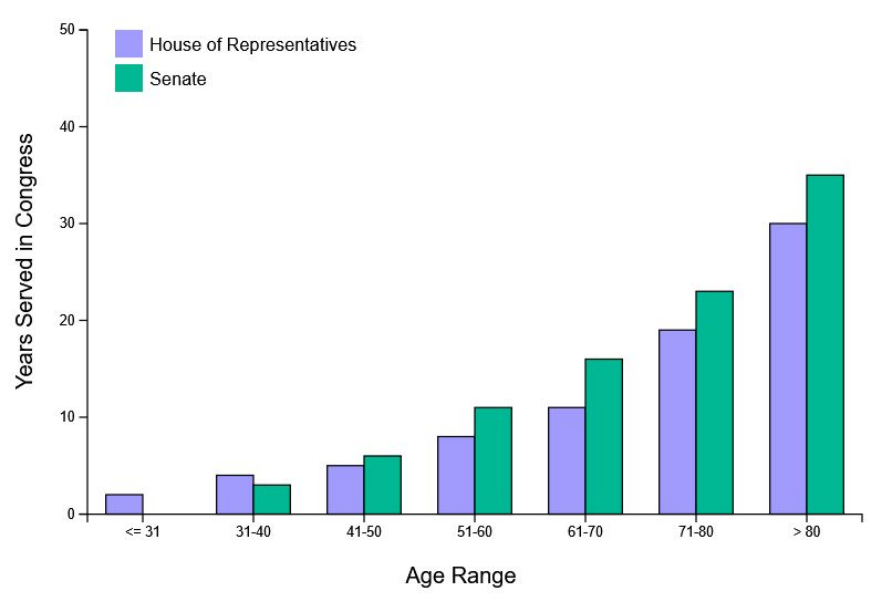
Design Evolution: Years Served Clustered Bar Chart

After creating the stacked bar chart in D3, we were successfully able to show the averages for each age range, but we ran into a problem. Because the Senate consists 100 members while the House consists of 435 members, many House members will have never served on the Senate. This means while some Senate members have served in the chamber for upwards of 35 years, that data point is undermined when averaged with the data of all the House members that year that have never served on the Senate once.



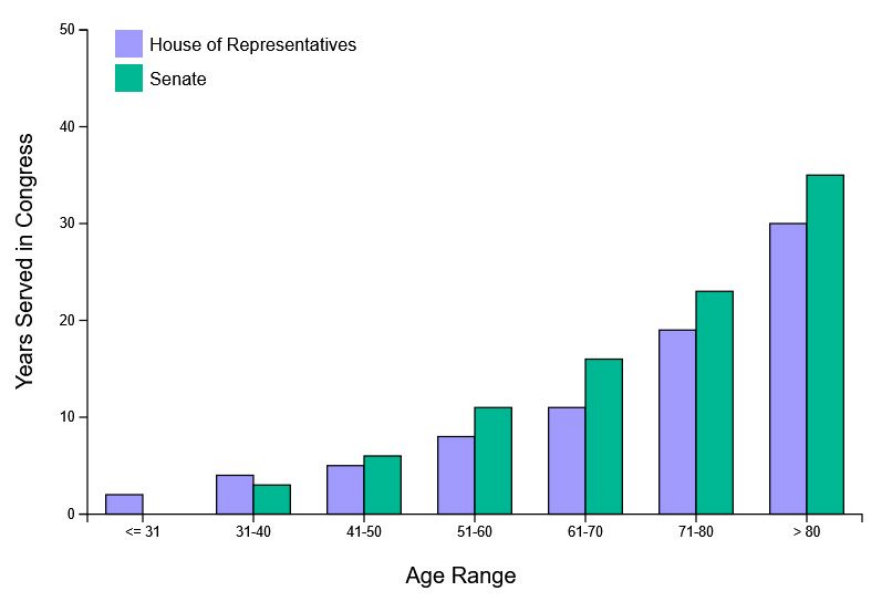
Design Evolution: Years Served Clustered Bar Chart

We realized that a stacked bar chart combining both chambers didn't display the data accurately, while two charts for experience for each chamber wouldn't be as useful, since House members tend to stay in the House and Senate members usually are only in the House a couple years before switching to the Senate.



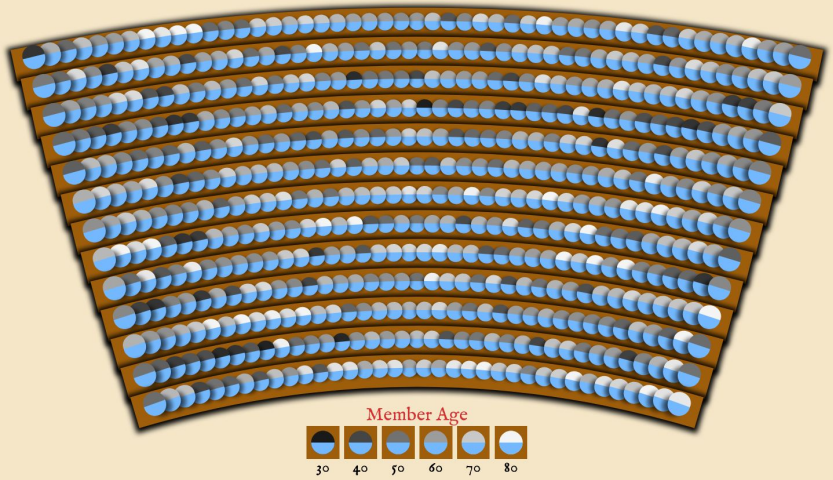
Design Evolution: Years Served Clustered Bar Chart

After some discussion, we decided to abandon showing the average member's time in the House versus the Senate and combine the bars into the average member's time in Congress as a whole. We then shifted to a clustered bar chart where each age range has a separate bar for the average of members in the House and the Senate. This meant that we could show how long members of both chambers have been in office during a specific year without the contrasting sizes of both chambers negatively affecting our visualization.



Implementation

Our icon graph is the first visualization that shows on our page and aims to be the eye-catcher for our site. The hair color of the icons gives a quick way to visualize the spread of ages present in Congress, and the resemblance to an actual chamber of Congress quickly clues readers in to what we intend to represent. This was one of the most work intensive tasks, as each icon allows a user to hover to see a person's age and year experience.



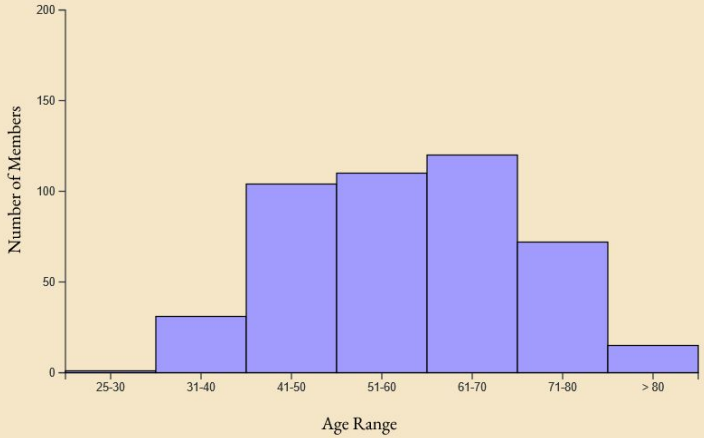
Implementation

Our bar graphs show the spread of ages in both chambers of Congress for a certain year. We split it up into the two chambers to illustrate if there's a difference in distribution between the House and the Senate or if the average age of Congress members is independent of chamber.

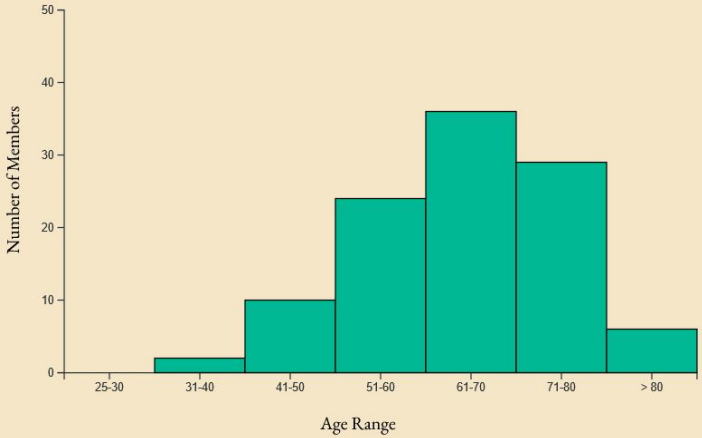
Since the House is 435 members while the Senate is 100 members, one thing to note about our visualizations is that the range differs between the two to make the differences between bars of one graph more apparent, and make general trends between graphs easier to compare.

Age Distribution of Congress

House of Representatives

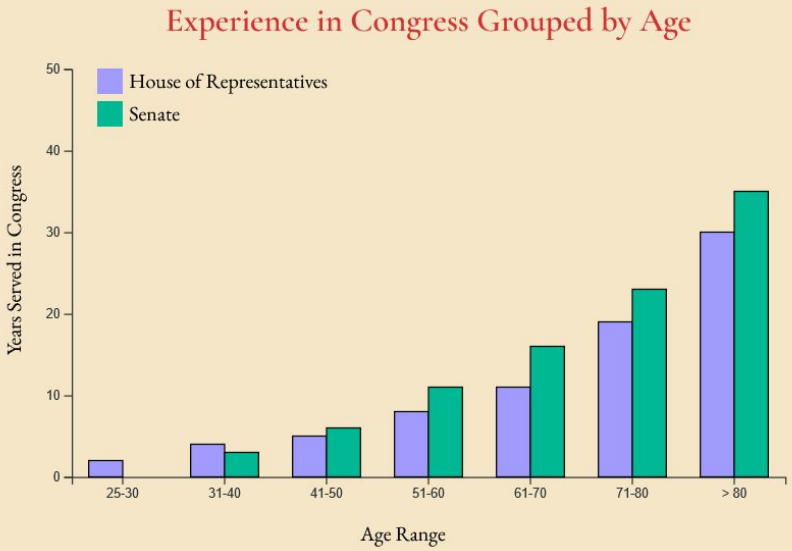


Senate

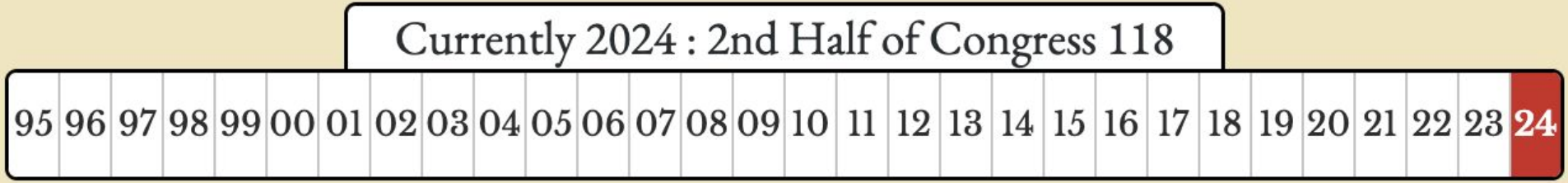


Implementation

The clustered bar chart groups its bars into the same ranges to make it intuitive to switch from the other bar charts to this one and apply the data from both to make a conclusion. The intent of this chart is to show the average years that members of Congress have served so far. Since a member can serve unlimited terms as long as they keep getting reelected, this graph illustrates how often members tend to stay in Congress once they're in.



Implementation



The main interactive element of our page is a timeline of all the years our visualization covers, in which a user can click a year and see the visualizations change to the congress of that year. This lets a user compare how the age representation in Congress has changed over the past 30 years and see trends in the age and experience distribution over time. When a user scrolls down, the timeline remains on the page, so they can switch years no matter where on the site they are.

Above the timeline, we included the year that's been selected, along with the associated Congress during that year for anyone that's curious.

Why Sort by Year?

While each Congress lasts two years, with House members' terms lasting two years and Senate members' terms lasting six years, we still decided to let users select a year to show a graph rather than the number of that year's congress. This means that the data for two subsequent years will be very similar beyond every member being one year older, and this makes what the data is saying a lot clearer to users. "This is the data for Congress in 2024" means a lot more to the average person than "This is the data for the 118th Congress", so we went with this choice to make for visualizations that readers can more easily gain something from. We still wanted to include the Congress number for a selected year, so we added it on top of the timeline for if readers are curious.

Evaluation

Our visualizations did end up helping answer the questions we propose and teaching us more about the ages of Congress members. One conclusion that we expected to see was that the age of Congress members leans towards the older side. Most members are in their 50s and 60s and there's a fair amount in their 70s and above. The ranges from 25-40 are consistently on the lower end, despite the fact that someone eligible can get elected for Congress as soon as they turn 25.

Evaluation

The age distribution of Congress members remains similar throughout the last 30 years, though there are some notable differences as time goes on. For one, the second highest age bracket for the House in the 90s was the 41-50 bracket, but after 2003, the 61-70 bracket passes it in members, and since then the amount of House members in their 40s has never exceeded the amount in their 60s. And while the age brackets tend to fluctuate, the number of members in the Senate over 70 is the highest it's been in the past 30 years. Both of these aspects signify a stagnating Congress as time goes on.

Evaluation

The clustered bar chart gives further insight into how long members remain in power. Since members on Congress can keep serving as long as they're reelected, and for every year you'll see members in their 70s and above have been serving for at least 20-30 years. It reveals that once someone in Congress has gotten into power, they have the opportunity to maintain it for decades.

Evaluation

If we were to further improve this site, we'd like to add a streamgraph visualization that would show the age distribution of Congress members over a number of years. While our interactive visualization does let you select years to compare one from another, letting users see the trends in Congress member ages from a glance would convey that information more clearly.

We'd also like to add a timeline visualization that shows how long the 10 oldest members have been serving and which historical events they've served during. This would offer a new perspective to how long some Congress members actually stay in office.